Diseases of the Cardiac Valves and Ischemic Heart Disease

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Structural valve dysfunction → Coronary artery disease → Angina → Myocardial infarction → Pericarditis, myocarditis, endocarditis → Summary

Three categories of cardiac pathophysiology are covered: structural valve dysfunction, coronary ischemic heart disease, and inflammatory or infectious conditions of the heart wall. As you study them, connect the mechanical disruptions to the foundational systemic concepts: fluid and electrolyte imbalance, acid-base balance, inflammation, immunity, and infection. Ask how altered pump mechanics drive fluid retention, pulmonary congestion, and lethal electrical instability; how ischemic tissue drives local and systemic lactic acidosis; and how cellular injury, autoimmune responses, and circulating pathogens damage delicate cardiac tissue.

Exam notes: There will be NO EKGs on the test this semester. Nitroglycerin and nursing interventions are covered next semester.

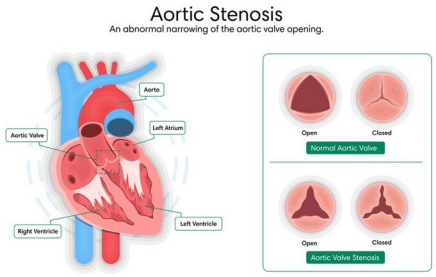
Structural Valve Dysfunction

Rather than memorizing each of the four valves individually (the lecture does not cover mitral stenosis specifically, for example), learn three mechanical concepts that govern all valve failures: stenosis, regurgitation, and prolapse.

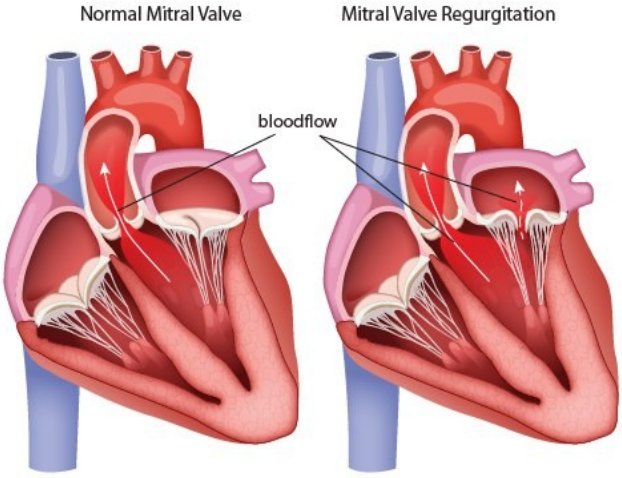
	Stenosis	Regurgitation (insufficiency)	Prolapse
Failure	Fails to open fully	Fails to close fully	Leaflet displacement
Mechanism	Leaflets become stiff, fused, or calcified, narrowing the orifice and obstructing forward flow when the valve should be open	Leaflets fail to seal, allowing backward (retrograde) leakage into the preceding chamber when the valve should be closed	Leaflets bulge backward into the preceding chamber during contraction, from weakened connective tissue or stretched chordae tendineae. Does not always impede flow, but if leaflets cannot meet completely it can cause regurgitation
Hemodynamic effect	PRESSURE overload in the chamber behind the valve: it pumps harder against high resistance, leading to compensatory hypertrophy	VOLUME overload in the chamber receiving the leaked blood, leading to progressive chamber dilation and eventually pump failure	Variable; leads to regurgitation if displacement prevents complete closure
Causes / risk factors	Age-related calcification (wear and tear); rheumatic heart disease; congenital abnormality; advanced age; hypercalcemia and chronic kidney disease accelerate calcification; males more than females	Infective endocarditis (direct valve destruction); connective tissue weakening; post-MI (cell death); dilated cardiomyopathy; hypertension (wall stress); connective tissue disorders; prior MI; IV drug use (infection)	Spongy, weakened connective tissue in the middle layer of the leaflet; genetic collagen mutations; connective tissue disorders; family history; most commonly young women

Rheumatic heart disease: post-streptococcal autoimmune cross-reactivity, seen after strep throat that was untreated (and sometimes even when treated).

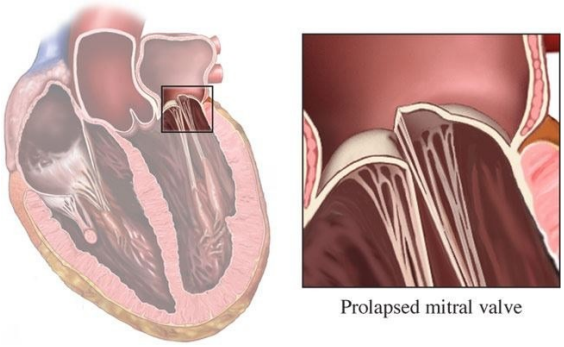
Example of stenosis: a stenotic mitral valve makes it harder for blood to flow from the left atrium into the left ventricle, so the chamber behind the valve must pump harder against that resistance. Example of regurgitation: when the left ventricle contracts, blood flows backward through the mitral (bicuspid) valve into the left atrium, which then holds excess volume it should not have.



Aortic stenosis (lecture slide)



Normal mitral valve vs. mitral regurgitation (lecture slide)



Prolapsed mitral valve (lecture slide)

Fluid Balance

Whether a valve is stenotic or regurgitant, the outcome is increased upstream hydrostatic pressure. Fluid backs up into the preceding chamber and capillary beds and is forced into the interstitial space, causing edema. Left-sided problems back blood up into the lungs (pulmonary congestion); right-sided problems cause peripheral edema throughout the body.

Diagnostics

- **Echocardiography (gold standard):** visualizes leaflet motion during contraction, measures pressure gradients across stenotic valves, quantifies the volume regurgitated, and calculates ejection fraction.
- **Cardiac catheterization:** gives precise intracardiac pressures across the valve orifices. More invasive, so not the first-line test.

- **BNP:** a hormone the heart releases when it works too hard. Evaluates wall stretch from volume or pressure overload and can be trended.

Echocardiograms and BNP levels are repeated periodically to trend progression and decide whether to intervene.

Coronary Artery Disease (CAD)

CAD is not a passive accumulation of fat clogging a pipe. It is an active, progressive, immune-mediated inflammatory disease of the arterial wall, specifically the tunica intima, that develops over decades.

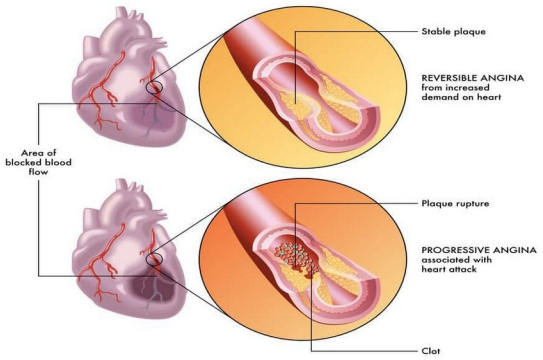
How a Healthy Vessel Becomes Disease-Vulnerable

1. **Endothelial dysfunction and injury:** normal endothelium produces nitric oxide (a potent vasodilator, anti-inflammatory, and inhibitor of platelet aggregation). Chronic exposure to stress, high blood pressure, toxins like cigarette smoke, and high glucose damages the lining and nitric oxide production drops. Vessels become more permeable, pro-inflammatory, and prone to vasospasm.
2. **Lipid infiltration and oxidation:** circulating LDL crosses into the subendothelial space of the intima. Exposed to reactive oxygen, trapped LDL is oxidized, which is highly cytotoxic and acts as the trigger for inflammation and the immune system.
3. **Leukocyte recruitment and foam cells:** damaged endothelial cells express adhesion molecules that capture circulating monocytes and helper T lymphocytes. Monocytes become macrophages with scavenger receptors that engulf oxidized LDL; engorged with lipid, their cytoplasm turns pale and they become foam cells. Foam cells collect beneath the endothelium as fatty streaks, the first lesions of atherosclerosis.
4. **Fibrous cap:** in response to macrophage cytokines, smooth muscle cells migrate from the tunica media into the intima and secrete collagen and elastin, forming a hard fibrous cap over the lipid deposits.
5. **Necrotic core:** beneath the cap, foam cells die (necrosis), releasing intracellular lipids, calcium, and tissue factors that form a soft, inflammatory necrotic core.
6. **Thrombus or rupture:** the plaque can build up a thrombus or rupture. This is where acute coronary syndromes begin.

Angina Pectoris

Chest pain caused by transient, reversible myocardial ischemia. It occurs when myocardial oxygen demand exceeds supply. Hypoxic myocardial cells switch from aerobic respiration to anaerobic glycolysis, producing lactic acid and releasing adenosine, which stimulates visceral sensory nerve fibers in the myocardium and causes pain.

Key point: If oxygen supply is restored within 20 minutes, cellular function recovers completely without permanent necrosis. Angina is NOT permanent ischemia: it comes and goes and is reversible. This is what separates it from MI.



Stable plaque (reversible angina) vs. plaque rupture and clot (progressive angina) (lecture slide)

	Stable angina	Unstable angina
Mechanism	Fixed atherosclerotic plaque narrows the coronary lumen by ≥ 70%, limiting oxygen supply during exertion or stress. Transient ischemia without lasting damage	Acute plaque rupture with superficial thrombus, or vasospasm, causing dynamic, partial occlusion that unpredictably reduces coronary flow
Pattern	Predictable and reproducible; occurs with exertion or emotional stress (exercise, walking, gardening)	Unpredictable; occurs at rest, increases in frequency or severity, or lasts longer than 15–20 minutes
Relief	Relieved by rest (or nitroglycerin, which you learn next semester)	Medical emergency with high risk of progressing to MI
Symptoms	Retrosternal chest pain: tightness, squeezing, heavy or burning ("an elephant sitting on my chest"). May radiate to left shoulder, arm, neck, jaw, or epigastrium. Mild dyspnea, mild diaphoresis, anxiety, nausea during exertion	Similar quality but often more severe and intense retrosternal pressure (though mild pain does not rule it out). Diaphoresis, nausea, vomiting, dyspnea, intense apprehension from the sympathetic surge
Hemodynamics	Patients typically stable at rest	Can significantly impair oxygen supply and cardiac output
Extra risk markers		Elevated baseline CRP, hypercoagulability, increased inflammatory state

Risk Factors and Triggers

- **Non-modifiable:** advanced age (a big trend across cardiac disease); male sex, with more females affected post-menopause; family history of early CAD.
- **Modifiable:** dyslipidemia (high LDL, low HDL), hypertension, cigarette smoking, diabetes mellitus (elevated glucose damages arteries), obesity, sedentary lifestyle.

- **Triggers that surge myocardial demand:** physical exertion, emotional stress, extreme cold, heavy meals.

Assessment tip: Ask open-ended questions about what the chest feels like, not just "Are you having pain, yes or no?" A patient may say no to pain but describe something squeezing. Get a thorough description of what it actually feels like.

Acid-Base Imbalance

Local tissue hypoxia shifts cells to anaerobic metabolism, generating hydrogen ions and lactic acid. This localized metabolic acidosis impairs the cardiac cell membrane sodium-potassium ATPase pump and destabilizes the resting membrane potential, predisposing the heart to dysrhythmias even before cell death occurs. Blood gases are not typically drawn for these patients; know that the impaired pump (with potassium effects on cardiac muscle) raises dysrhythmia risk.

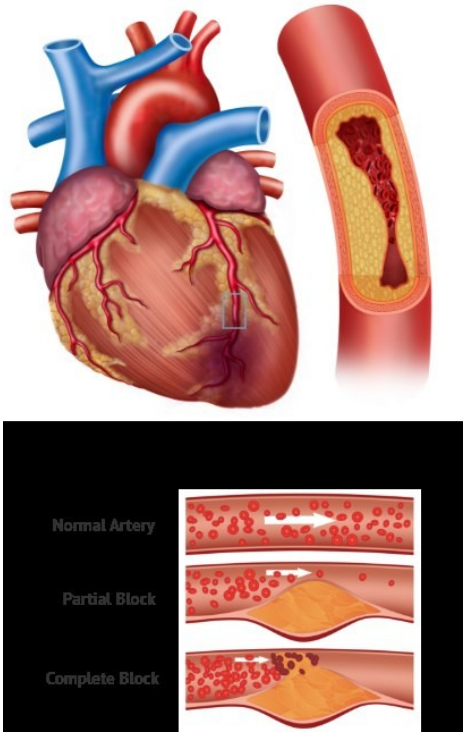
Diagnostics

- **12-lead EKG:** patients with an MI may arrive not knowing whether it is angina or MI, so the EKG helps rule out MI.
- **Cardiac biomarkers (troponin and CK-MB):** NORMAL in stable and unstable angina. An elevated troponin means myocardial necrosis has occurred, which defines an MI. This is the key test separating angina from MI.
- **Exercise stress test:** evaluates ischemic EKG changes, especially for stable angina; done outpatient, not right away.
- **Coronary CT angiography (CCTA):** directly visualizes plaque burden and lumen stenosis.

Both forms affect patients physically and emotionally; they must stay aware of recurrence, and lifestyle modification for prevention is essential.

Myocardial Infarction (MI)

A heart attack. When myocardial ischemia persists beyond 20 minutes, reversible injury transforms into irreversible myocardial death. The process is much like angina, but far more dangerous. Plaque rupture or thrombosis exposes subendothelial collagen, platelets aggregate and activate the coagulation cascade, and a thrombus completely or partially occludes the coronary artery.



Coronary artery occlusion in MI (lecture slide)

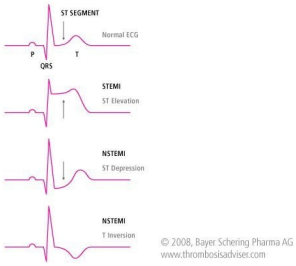
Ischemic Timeline

When	What happens
Seconds	Aerobic metabolism stops; ATP stores deplete.
Within 20 minutes	Irreversible cell damage (coagulative necrosis) begins in the subendocardium, the most vulnerable layer, and progresses outward toward the epicardium.
24–48 hours	Neutrophils infiltrate the necrotic tissue and start inflammatory clearance.
2–4 weeks	Macrophages replace necrotic myocardium with non-contractile collagenous scar, causing altered ventricular wall motion and remodeling.

STEMI vs. NSTEMI

Exam note: No EKGs on the test. This is to help you understand what we look for when diagnosing.

	NSTEMI	STEMI
Occlusion	Partial or transient occlusion	Complete transmural occlusion
Damage	Subendocardial infarction (inner layer of the wall only)	Full-thickness infarction of the myocardial wall
EKG	ST depression or T wave inversion	ST segment ELEVATION: the hallmark sign of MI



Normal EKG vs. STEMI vs. NSTEMI patterns (lecture slide)

Risk Factors and Triggers

- **Non-modifiable:** advanced age, male sex, post-menopausal female status, family history.
- **Modifiable:** dyslipidemia, hypertension, cigarette smoking, diabetes, obesity, sedentary lifestyle.
- **Acute triggers (in those with other risk factors):** a big sympathetic surge, severe hypercoagulable state, medication toxicity, cocaine or other stimulants, systemic infection or sepsis.

Signs and Symptoms

Hallmark: severe, crushing substernal pressure or heaviness radiating to the left arm, jaw, neck, or back. Autonomic symptoms from the sympathetic surge: diaphoresis, cold clammy skin, nausea, vomiting, profound apprehension, possible palpitations. Most are the same as in angina.

	Males	Females
Typical blockage	More often large coronary arteries	More often the smaller arteries that branch off the coronary arteries
Presentation	Stereotypical: sudden cold sweat, chest pain and pressure, discomfort or tingling in back, shoulders, arms, neck, jaw	Neck or jaw pain often without chest pain, upper back pain, nausea and vomiting, extreme fatigue, heartburn or indigestion, lightheadedness or dizziness

Because women often lack the hallmark chest pain, their symptoms are not recognized, they seek care later, and outcomes tend to be worse. (The lecture links an American Heart Association video about a young woman, a healthcare worker, who delayed care during an MI.)



Signs of myocardial infarction (lecture slide)

Diagnostics

Anyone who comes in with chest pain gets an EKG, and troponin and CK-MB are drawn to rule out an MI.

Test	Finding / timing
12-lead EKG	ST elevation; ST depression or T wave inversion
Troponin	Starts to rise 2–4 hours after injury (may still look normal at first); peaks around 18–24 hours; stays elevated 7–14 days. Trended on some patients to follow myocardial necrosis
CK-MB	Specific cardiac enzyme marker: rises within 4–6 hours and normalizes in 40–72 hours. Useful for detecting early reinfarction
CBC	Leukocytosis, peaking about 2–4 days after the MI, from systemic inflammatory clearing of necrotic tissue
Echocardiogram	Identifies regional wall motion abnormalities where the infarct was; good for follow-up, not done immediately

After an MI, patients are at higher risk for emboli, thrombus formation, arrhythmias, cardiac failure, shock, and myocardial dysfunction because the heart walls are permanently damaged, so adherence to the treatment plan is essential.

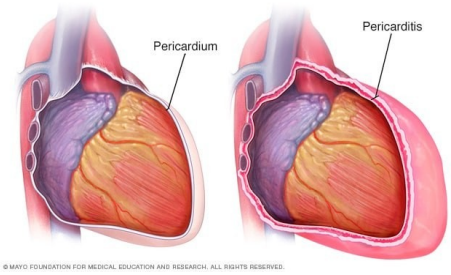
Inflammatory and Infectious Heart Disorders

Each condition affects a different layer of the heart. Match the name to the layer you learned earlier:

- **Pericarditis:** the pericardium, the outer sac.
- **Myocarditis:** the myocardium, the middle muscle layer.

- **Endocarditis:** the endocardium, the inner lining.

Acute Pericarditis



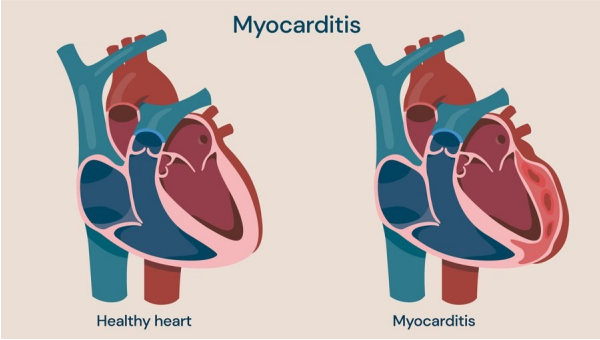
Normal pericardium vs. pericarditis (lecture slide)

Inflammation of the pericardial sac. Cytokines cause pericardial vasodilation, hyperpermeability, and exudate accumulation, producing friction between the parietal and visceral layers. Cytokines also lead to prostaglandin production, which increases the inflammation. Fluid can accumulate in the sac (pericardial effusion), like a pleural effusion around the lungs.

- **Causes:** viral infection (most common), post-MI inflammation, autoimmune disorders (lupus, rheumatoid arthritis), uremia from kidney failure.
- **Hallmark symptom:** sharp, pleuritic chest pain that worsens on inspiration and when lying flat, and gets better sitting up and leaning forward.
- **Pericardial friction rub:** high-pitched scratching or squeaking, heard best at the left lower sternal border with the patient leaning forward. Considered a hallmark but rarely heard.
- **EKG:** ST segment elevation with PR segment depression.
- **Labs:** elevated ESR and CRP (systemic inflammation). Troponin should NOT be elevated.

Pericarditis or MI? Both can present with chest pain and ST elevation. Clues pointing to pericarditis: normal troponin, and a patient without MI risk factors (fit, non-smoker, no MI history) who had a viral illness a couple of weeks ago. Treatment differs: these patients rest until cardiac enzymes and inflammation stabilize, then return to normal activity.

Myocarditis

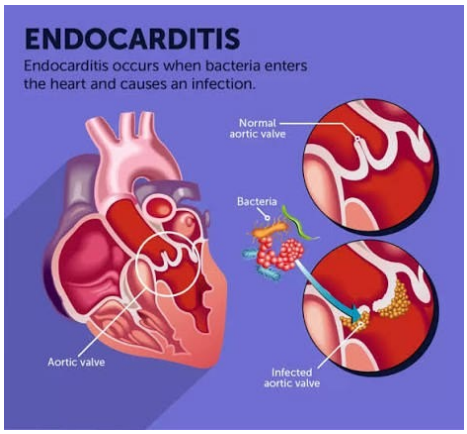


Healthy heart vs. myocarditis (lecture slide)

Inflammation of the myocardial muscle layer. Inflammatory cells infiltrate and cause myocyte necrosis, cellular edema, and loss of contractile strength, often progressing to dilated cardiomyopathy or acute heart failure (next lecture).

- **Development:** myocardial injury (most often viral infection; also autoimmune disease, toxins, or medications) → immune response in which cytokines and T cells mistakenly attack cardiac cells → tissue damage and necrosis → either resolution or progression to chronic myocarditis with lasting dysfunction.
- **Etiology:** direct viral invasion (e.g., COVID-19, adenovirus), autoimmune activation, cardiotoxins (adverse effects of medications).
- **Signs and symptoms:** fatigue, dyspnea, palpitations, signs of acute heart failure (edema, S3 gallop), unpredictable dysrhythmias from inflammation affecting conduction, chest pain.
- **Diagnostics:** elevated troponin and CK-MB; elevated ESR/CRP; endomyocardial biopsy is the gold standard to confirm.
- **Impact:** ranges from full recovery to long-term heart failure and a chronic condition to manage.

Infective Endocarditis



Normal vs. infected aortic valve (lecture slide)

Inflammation of the lining and chambers of the heart, most commonly damaging the heart valves. Endothelial injury (often from turbulent flow or procedures) lets pathogens from bacteremia adhere to cardiac tissue and form vegetations. In infective endocarditis these contain bacteria, viruses, or fungi; in non-infective endocarditis the vegetations are sterile, made of platelets and fibrin. It can be fatal if pieces reach the bloodstream.

Risk Factors

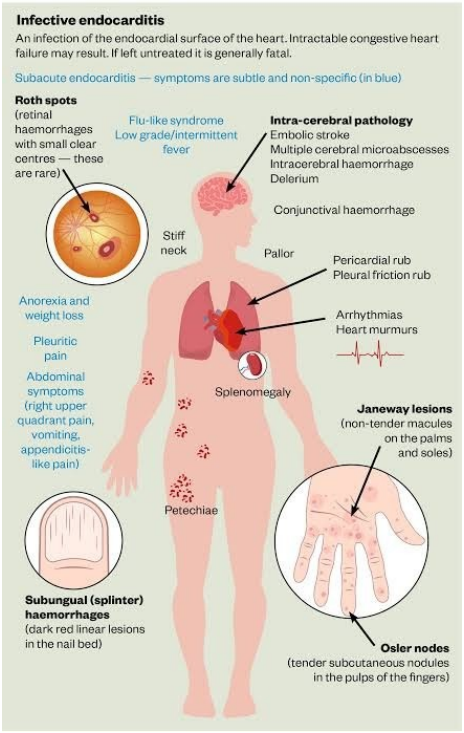
- **IV drug use:** introduces bloodstream infection (unclean injection sites, frequent injection).
- **Central line catheters:** a hole in the body that microorganisms can enter (hence the focus on preventing catheter-related bloodstream infections).
- **Prosthetic heart valves:** easy for organisms to adhere to.
- **Structural heart disease**
- **Recent invasive dental or surgical procedures:** a common cause; teeth cleaning or invasive dental work can introduce oral bacteria into the bloodstream, so ask about recent dental visits. High-risk patients (prosthetic valves, prior infective endocarditis, existing cardiac problems) still get dental care but receive prophylactic antibiotics beforehand. Patients without these risks do not, so it still occurs.

Signs and Symptoms

- **Non-specific:** fever, fatigue, night sweats, muscle aches, anorexia.

- **Specific (patients rarely mention them):** new or changing heart murmur; splinter hemorrhages (dark longitudinal streaks under the nails); petechiae (tiny flat pinpoint red, brown, or purple spots from broken capillaries); Janeway lesions (painless erythematous macules on palms or soles); Osler nodes (painful, tender raised nodules on finger and toe pads); Roth spots (retinal hemorrhages with pale centers). Chronic: splenomegaly, finger clubbing.

Why they occur: Petechiae, Osler nodes, Roth spots, and splinter hemorrhages come from small-vessel damage from inflammation and immune reaction. Janeway lesions are tiny clumps of bacteria or cells shed from the valve that block small vessels, causing painless spots without major inflammation. Not every patient has all of them; some have none.



Signs of infective endocarditis (lecture slide)

Diagnostics

- **Blood cultures (gold standard):** at least two, preferably three, from independent venipuncture sites.
- **CBC:** white blood cell count.
- **Echocardiogram:** shows valvular vegetations, abscesses, or a new regurgitant jet; guides the plan of care.

Complications depend on whether the infection stays in the heart or spreads: heart failure, arrhythmias, embolism, pneumonia, stroke, and poor perfusion elsewhere from emboli. Preexisting conditions such as implanted valves also affect outcomes.

Summary and Quick Review

- Stenosis causes pressure overload. Regurgitation causes volume overload. Both drive fluid backward into the pulmonary or systemic circulation (think forward-flow and backward-flow effects).

- Angina is reversible ischemia with normal troponin. MI is irreversible cell death with elevated troponin.
- Acidosis and electrolyte shifts (potassium release) threaten electrical stability.
- Pericarditis: pleuritic pain relieved by leaning forward, with ST elevation.
- Myocarditis: weakens muscle contraction (inflammation or infection of the muscle itself).
- Endocarditis: infected vegetations that destroy valves and can cast septic emboli throughout the body.

	Angina	MI	Pericarditis
Mechanism	Reversible ischemia (demand > supply)	Irreversible necrosis; ischemia > 20 min	Inflammation of the pericardial sac
Pain	Exertional (stable) or unpredictable (unstable); eased by rest in stable	Severe, crushing, radiating; autonomic symptoms	Sharp, pleuritic; worse inspiring or lying flat, better leaning forward
Troponin / CK-MB	Normal	Elevated	Not elevated
EKG	Ischemic changes possible	ST elevation, or ST depression / T inversion	ST elevation with PR depression

	Myocarditis	Endocarditis
Layer	Myocardium	Endocardium (usually valves)
Key features	Fatigue, dyspnea, palpitations, HF signs, dysrhythmias; troponin elevated	Fever, new murmur, splinter hemorrhages, Janeway lesions, Osler nodes, Roth spots
Gold standard	Endomyocardial biopsy	Blood cultures (2–3 sites)

Since many of these conditions share symptoms, use the diagnostic tests as cues: anticipate what the provider will order and know what to look for when results return.

Notes on Transcript Cleanup

Corrected mis-transcriptions: "epithelial" is endothelial; "infraction" is infarction; "coronary artery syndrome" is acute coronary syndrome; "erythrocyte sedation rate" is sedimentation rate; "erythrocytes and CRPs" is ESR and CRP; "retorretinal" is retinal. In the stenosis example the speaker said the "right atrium" must pump harder, but for the mitral valve example it is the left atrium. She also said the subendocardium is the vulnerable layer "of our coronary arteries"; it is a layer of the myocardium. The line about "COVID-19, we call this adenovirus" is garbled, so I listed both as examples of viral causes.